Successful ECT treatment for medically refractory nonconvulsive status epilepticus in pediatric patient.

PMID: 21333551 | Indexed in PubMed

Status epilepticus is a life threatening condition with a high mortality rate in spite of aggressive treatment. There is little consensus on third and fourth line approaches in refractory cases. While electroconvulsive therapy (ECT) has been employed successfully as a treatment for refractory epilepsy and status epilepticus (SE) after exhausting conventional therapy, its use for pediatric patients is limited. We describe a 7-year-old pediatric case in which ECT was used successfully to treat medically refractory nonconvulsive status epilepticus (NCSE) without complete withdrawal of antiepileptic drugs (AED).

ECT in the treatment of status epilepticus.

PMID: 11528315 | Indexed in PubMed

Owing to its potent anticonvulsant actions, electroconvulsive therapy (ECT) has been proposed as an intervention for treatment-resistant seizure disorders. We review the literature on the use of ECT in treatment-resistant epilepsy and status epilepticus (SE) and present a case of a patient who was in nonconvulsive SE for 26 days and then treated with ECT after all standard pharmacological strategies were exhausted. Because of skull defects, a novel electrode placement was used. Owing to massively elevated seizure threshold attributable to concomitant anticonvulsant medications, extraordinarily high electrical dosage was needed for ECT to elicit generalized seizures. Status was terminated after three successful ECT-induced seizures. However, the long-term functional outcome of the patient was poor. The role of ECT in the treatment algorithm for SE is discussed.

Maintenance ECT as a therapeutic approach to medication-refractory epilepsy in an adult with mental retardation: case report and review of literature.

PMID: 22531207 | Indexed in PubMed

Electroconvulsive therapy (ECT) raises the seizure threshold. This physiological change may benefit patients with seizure disorders. Whereas ECT has recently been used to terminate medication-refractory status epilepticus, there is little current literature on its planned administration as a specific maintenance treatment for medication-refractory epilepsy. We used maintenance ECT to treat an 18-year-old man with a long-standing generalized tonic-clonic seizure disorder who had shown poor response to several antiepileptic drugs administered in combination with antiepileptic medication compliance confirmed through drug level monitoring. A total of 52 ECTs were administered across nearly 20 months at a mean frequency of once in nearly 12 days. From the very outset, ECT dramatically decreased the frequency of spontaneous seizures from approximately 6 to 24 per week at baseline to approximately 1 to 2 per week after ECT initiation. The efficacy of maintenance ECT in spontaneous seizure prophylaxis was greater when the ECT treatment interval was narrower. Improvement with ECT was associated with improved behavior and improved psychosocial functioning on clinical report. No cognitive or other adverse effects were reported or clinically ascertained. The ECT charge administered at the last 10 treatment sessions was 1434 millicoulombs. This is probably the highest electrical stimulus dose recorded in literature. Maintenance ECT may reduce the frequency of breakthrough seizures in patients with seizure disorder that is inadequately responsive to antiepileptic medication regimes. Very high ECT seizure thresholds may be observed when many antiepileptic drugs are concurrently administered in high doses.

Electroconvulsive Therapy in Super Refractory Status Epilepticus: Case Series with a Defined Protocol.

PMID: 32516983 | Indexed in PubMed

Super-refractory status epilepticus (SRSE) represents a neurological emergency that is characterized by a lack of response to the third line of antiepileptic treatment, including intravenous general anesthetics. It is a medical challenge with high morbidity and mortality. Electroconvulsive therapy (ECT) has been recommended as a nonpharmacologic option of treatment after other alternatives are unsuccessful. Its effect on the cessation of SRSE has been minimally investigated. The objective of this article is to analyze the effect of ECT on SRSE. For this purpose, a multidisciplinary team created a protocol based on clinical guidelines similar to those described previously by Ray et al. (2017). ECT was applied to six patients with SRSE after the failure of antiepileptic treatment and pharmacologic coma.The objective of each ECT session was to elicit a motor seizure for at least 20 s. SRSE was resolved in all patients after several days of treatment, including ECT as a therapy, without relevant adverse effects. Thus, ECT is an effective and feasible option in the treatment of SRSE, and its place in the algorithm in treatment should be studied due to the uncommon adverse effects and the noninvasive character of the therapy.

Treatment of status epilepticus with electroconvulsive therapy.

PMID: 17435572 | Indexed in PubMed

To describe a case in which electroconvulsive therapy (ECT) was used successfully to treat refractory status epilepticus (SE) after all pharmacological therapies were exhausted. A 39-year-old man with no seizure history presented in SE secondary to presumed viral encephalitis. His seizures remained refractory to medical management, and he was placed in a pentobarbital-induced coma. Multiple attempts to wean pentobarbital over the next several months failed due to SE relapses. With all standard pharmacological therapies exhausted, the patient underwent a series of 3 ECT sessions per day for 3 consecutive days. Electroencephalogram improvements were noted immediately with diffusely slow activity and with a delayed response over time in the patient's neurological examination. Twelve months post-ECT, the patient is awake, alert, and being managed on antiepileptic medications as outpatient. This case further illustrates the role of ECT in the treatment of refractory SE.

Electroconvulsive therapy for super refractory status epilepticus in pregnancy: case report and review of literature.

PMID: 35287528 | Indexed in PubMed

We aim to describe use of electroconvulsive therapy (ECT) to treat super refractory status epilepticus (SRSE) in pregnancy and review the literature regarding utility and safety of ECT in refractory status epilepticus. Status epilepticus (SE) is a commonly encountered emergency in neuro-critical care world.

Pharmacotherapy of status epilepticus in pregnancy is very challenging given the effect of the majority of antiepileptic drugs (AEDs) on fetal development. Although there has been growing evidence for use of ECT in status epilepticus, data about its utility in pregnancy is lacking. A twenty-one year old Caucasian female with history of epilepsy presented at 8 weeks of gestation as status epilepticus (SE) after abrupt discontinuation of her AEDs. Treatment was initiated with standard regimen of benzodiazepine and levetiracetam, which was progressively expanded to include approximately 10 anti-epileptic drugs over the course of 30 days. The status epilepticus was super refractory to sedation. She underwent ECT on day 31 with remarkable improvement in electroencephalogram (EEG) pattern and resolution of status epilepticus following a single ECT session. We reviewed PubMed and collated case reports involving the use of ECT in status epilepticus with emphasis on differences in various confounding factors esp. etiology of status and age group. Our case is the first reported case of ECT for successful treatment of SRSE in pregnancy. While majority AEDs pose a significant maternal and fetal risk during pregnancy, ECT could be a potential frontline therapy for SE in pregnancy.

Pulmonary function tests for evaluating the severity of Duchenne muscular dystrophy disease.

PMID: 36596294 | Indexed in PubMed

In Duchenne muscular dystrophy (DMD), lung disease contributes significantly to morbidity and mortality. This study aimed to assess the usefulness of various pulmonary function tests in evaluating DMD severity. This retrospective study analysed lung function tests of patients with DMD-treated in the multidisciplinary respiratory neuromuscular clinic at Schneiders' Children Medical Center of Israel. Data were analysed according to age, ambulatory status and glucocorticoid treatment. Among 90 patients with DMD, 40/63 (63.5%) ambulatory patients and 22/27 (81.5%) nonambulatory patients successfully performed spirometry. Significant annual declines were demonstrated among nonambulatory patients, in percentile predicted forced vital capacity (3.8%) and in total lung capacity (5.5%) per year. The decline correlated with age and loss of ambulation but not with steroid treatment. Peak cough flow values were randomly distributed and did not correlate with age, ambulation or treatment. In nonambulatory patients, transcutaneous carbon dioxide measurement correlated significantly with age ($r = 0.55$, $p = 0.02$). Forced vital capacity, total lung capacity and transcutaneous carbon dioxide correlated with the clinical severity of disease in children with DMD. These measures may be useful in follow-up and clinical trials. A comparable correlation was not found for peak cough flow.

Measurement of respiratory function decline in patients with Duchenne muscular dystrophy: a conjoint analysis.

PMID: 29412787 | Indexed in PubMed

In Duchenne muscular dystrophy (DMD), little attention has been paid to severity of respiratory function decline (RFD) based on disease progression. We performed a conjoint analysis among 123 Italian clinicians to generate a scale for RFD in DMD patients. Before the interview, 11 attributes were selected by discussion among experts. 32 'patient profiles' were generated. Each physician assessed the severity of RFD for each profile. Each level/attribute was assigned an estimated usefulness to understand its impact on RFD. The identified attributes were forced vital capacity, forced vital capacity decline, dysphagia, type of ventilation and peak cough flow. These results allowed the development of a scale for RFD severity. This scale can stratify DMD patients according to the severity of their RFD.

Respiratory muscle decline in Duchenne muscular dystrophy.

PMID: 23836708 | Indexed in PubMed

Duchenne muscular dystrophy (DMD) causes progressive respiratory muscle weakness. The aim of the study was to analyze the trend of a large number of respiratory parameters to gain further information on the course of the disease. Retrospective study. 48 boys with DMD, age range between 6 and 19 year old, who were followed in our multidisciplinary neuromuscular clinic between 2001 and 2011. Lung function, blood gases, respiratory mechanics, and muscle strength were measured during routine follow-up over a 10-year period. Only data from patients with at least two measurements were retained. The data of 28 patients were considered for analysis. Four parameters showed an important decline with age. Gastric pressure during cough (Pgas cough) was below normal in all patients with a mean decline of 5.7 ± 3.8 cmH2 O/year. Sniff nasal inspiratory pressure (SNIP) tended to increase first followed by a rapid decline (mean decrease 4.8 ± 4.9 cmH2 O; 5.2 ± 4.4% predicted/year). Absolute forced vital capacity (FVC) values peaked around the age of 13-14 years and remained mainly over 1 L but predicted values showed a mean 4.1 ± 4.4% decline/year. Diaphragmatic tension-time index (TTdi) increased above normal values after the age of 14 years with a mean increase of 0.04 ± 0.04 point/year. This study confirms the previous findings that FVC and SNIP are among the most important parameters to monitor the evolution of DMD. Expiratory muscle strength, assessed by Pgas cough, and the endurance index, TTdi, which are reported for the first time in a large cohort, appeared to be informative too, even though measured through an invasive method.

Duchenne muscular dystrophy respiratory profiles from real world registry data.

PMID: 37539841 | Indexed in PubMed

Understanding real-world profiles from neuromuscular databases is helpful for optimizing clinical care and planning research studies. The Canadian Neuromuscular Disease Registry (CNDR) has respiratory data from a population of boys with Duchenne Muscular Dystrophy (DMD). To describe cross-sectional respiratory profiles from a national DMD real-world dataset. To explore the relationship between forced vital capacity percent predicted (FVC%) and disease severity parameters: scoliosis, ambulation and ventilation status. Descriptive statistics summarized the respiratory profiles. The CNDR registry enrolls and collects DMD clinic data from 36 Canadian centers. There were 414 participants enrolled. The age ranged from 2 to 36 years old. Pulmonary function test data were available for 323 participants. The use of ventilatory support was seen in a significant proportion (19.5%) of subjects by age 14-16 years and was used by the majority (69.2%) by age 20-22 years. FVC% declined at a rate of 3.19% per year with every 1-year increase in age. FVC% declined annually by 2.47% in nonambulatory participants versus by 0.96% in ambulatory participants. FVC% did not significantly change over age with the presence of scoliosis or use of ventilatory technology. The data from this large cohort are valuable for understanding real-world patterns of clinical care and disease progression. There is a significant association between the loss of ambulation and the rate of FVC% decline. Further longitudinal studies are needed to better understand the impact of disease parameters on pulmonary function decline and the need for ventilatory support.